"Shelf Smart Library Management System Using RFID tags and IOT"

Prof. A.A.Chaphadkar,

Professor, Department of Electronics and telecommunication, Sinhgad Institute of Technology and Science, Narhe, Pune.

Mr. Sagar Gaikwad, Mr. Omkar Joshi, Mr. Sagar Kale, Mr. Vinit Kapure

UG Student, Department of Electronics and Telecommunication, Sinhgad Institute of Technology and Science, Narhe, Pune.

Abstract: Designed to automate book tracking and organization in departmental shelves using RFID technology and IoT. Each book is tagged with an RFID chip, and RFID readers track check-in/check-out activities. The system is enhanced with voice alerts to detect misplaced books, ensuring that only books belonging to specific departments are placed on designated shelves. The backend of the system is developed using Python, while the frontend utilizes HTML, CSS, and JavaScript. A web application displays book information, manages issuing records, and sends email reminders to borrowers about return deadlines. The project aims to enhance library efficiency, prevent book misplacement, and simplify book management.

INTRODUCTION

Overview of Project

Libraries are evolving with the introduction of technology to enhance efficiency in managing vast collections of books. Traditional systems for cataloging and tracking books often result in misplacements and delays in book returns. This paper introduces a smart library management system that utilizes RFID technology and IoT to streamline book tracking and department- specific organization. The system is divided into two modules: one focusing on RFID-based tracking for book check-in/check-out, and another focusing on ensuring that books are correctly placed in department-specific shelves, supported by a voice module that alerts users of incorrect placements. This paper will also discuss the software and hardware components used, including ESP328 microcontrollers, RFID readers, and a web-based application. The backend is powered by Python, with the frontend created using HTML, CSS, and JavaScript.

RFID TECHNOLOGY AND ITS HISTORY :

RFID (Radio Frequency Identification) is a wireless sensing technology that relies on detecting electromagnetic waves. A standard RFID system is composed of three key elements: a transponder or tag, an interrogator, and a controller. The interrogator collects data, which is then sent to the controller through a network interface. The controller processes this data for various applications.

RFID tags operate across different frequency ranges, categorized into low frequency (LF), high frequency (HF), ultra-high frequency (UHF), and microwave. HF passive tags are widely utilized in libraries and bookstores for managing book inventories and controlling access. However, UHF passive tags have recently gained traction in library systems. UHF offers notable advantages over HF, including faster reading speeds and a broader read range. This enables quicker detection of multiple items and enhances efficiency in inventory management.

Despite these advantages, using UHF RFID in library self-service systems requires careful wave control to prevent misreading unintended items in close proximity, which could lead to errors in check-in and check-out processes. The origins of RFID technology can be traced back to 1939.

LITERATURE SURVEY

An RFID-Based Library Management System uses RFID tags on books and RFID readers at check-in/out stations and library exits to streamline operations. It enables quick, contactless check-ins/outs, enhances inventory tracking, and provides anti-theft security. This system improves efficiency, accuracy, and user convenience by automating processes. Staff can locate books easily, conduct faster inventory audits, and focus on other tasks. [1].

A Web-based Library Management System is an online platform that allows users to search for, reserve, and borrow books from any device. It provides an interface for librarians to manage book inventories, user accounts, and loan records efficiently. With user logins, members can view their borrowing history and due dates. The system offers realtime catalog updates and automated notifications for reminders and alerts. Accessible 24/7, it enhances user convenience and streamlines library operations. [2].

An RFID-Based Library Management System uses RFID tags on books and RFID readers to streamline book tracking, check-in/checkout, and inventory management. It enables quick, contactless operations, allowing users to borrow and return items easily. RFID readers at exits enhance security by detecting unauthorized removals. This system reduces manual workload, improves accuracy, and provides real-time updates on book locations. Overall, it enhances efficiency and convenience for both library staff and users. [3].

An Intelligent Library Management System using GSM and RFID automates book tracking and notifications. RFID enables quick, contactless check-in/check-out and real-time inventory updates, while GSM sends reminders or overdue alerts directly to users' phones. This system boosts security and reduces manual work for library staff. It enhances efficiency and keeps users informed." [4].

An IoT-Based Smart Library System with NFC allows quick book check-in/out via NFC tags and provides real-time inventory tracking. IoT integration monitors book locations and syncs data across devices. Security features prevent unauthorized removals, while alerts keep users updated. This system enhances efficiency and convenience.. [5].

A DSS for Library Management using RFID helps optimize book tracking and inventory management. It provides real-time data on book usage, availability, and location. The system supports decisionmaking with analytical insights for resource allocation. This improves operational efficiency and library management. [6].

The Smart Library Seat, Occupant, and Occupancy Information System uses pressure and RFID sensors to monitor seat availability and usage. Pressure sensors detect if a seat is occupied, while RFID tags identify the user. The system provides real-time occupancy data, improving space management. It enhances the user experience by optimizing seat allocation and tracking library space usage. [7].

A Library Management System using RFID Technology automates book tracking and inventory management. RFID tags on books enable quick check-in/check-out and real-time location tracking. The system improves security by detecting unauthorized book removals at exits. It enhances operational efficiency, reducing manual tasks and providing accurate data for librarians. [8].

An IoT-Based Secured Smart Library System with NFC-based book tracking uses NFC tags for quick and easy check-in/check-out. IoT integration provides real-time book location tracking and data synchronization across devices. The system enhances security by preventing unauthorized book removals. It improves library management efficiency and user experience through automated alerts and updates. [9].

Problem Statement

Libraries struggle with tracking books and managing check-outs manually, leading to errors and inefficiencies. Our project will use RFID and IoT technology to automate these tasks, making the system more accurate and efficient.

Objectives

Automate sorting and tracking of misplaced books using RFID.
Ensure smooth motion of the RFID reader for accurate data capture.
Enable real-time item location monitoring.
Reduce manual effort in re-shelving books.
Improve efficiency in locating and restoring misplaced books.

METHODOLOGY

Requirements Analysis:

The Shelf Smart Library Management System using RFID tags and IoT integration requires RFID tags, readers, IoT sensors, self-service kiosks, and security gates for automating book tracking and management. Software includes an integrated Library Management System, IoT platform, mobile app, and analytics dashboard. It must deliver an intuitive experience for users and efficient tools for librarians and administrators. Scalability, reliability, and security are key non-functional requirements. The system aims to streamline operations and enhance user engagement in libraries.

Model Design:

The model design of the Shelf Smart Library Management System using RFID tags and IoT integration includes RFID tags on books, with RFID readers and IoT sensors for real-time tracking and shelf monitoring. The centralized Library Management System (LMS) processes data and manages inventory, integrated with self-service kiosks for check-out and return. A mobile app allows users to interact with the system, including reservations and notifications. Cloudbased infrastructure enables remote management and data storage. An analytics dashboard provides insights for library administration and decision-making.

Intelligent Access Control:

Intelligent access control in the **Shelf Smart Library Management System using RFID tags and IoT integration** uses RFID security gates to prevent unauthorized removal of books. The system ensures that only checked-out books pass through gates, with real-time alerts for suspicious activity. IoT sensors on shelves monitor book placement and detect any tampering. User access is granted via RFID tags or the mobile app for check-out, ensuring secure interactions. Data from access events is logged and analyzed for security insights and improved management.

Integration and Testing:

Integration of the **Shelf Smart Library Management System using RFID tags and IoT integration** involves connecting RFID readers,
IoT sensors, kiosks, and the mobile app to the central Library Management System (LMS) for seamless communication. Testing includes verifying the accuracy of RFID tracking, shelf monitoring, and security features. System performance, user interfaces, and mobile app functionality are tested for usability and reliability.

Deployment:

Deployment of the Shelf Smart Library Management System using RFID tags and IoT integration involves setting up RFID infrastructure, IoT sensors, and self-service kiosks at library locations. The system is installed on cloud-based servers for remote management and data storage. User training is provided, and the system is fully operational after testing and configuration

Evaluation:

The evolution of the Shelf Smart Library Management System using RFID tags and IoT integration moves from manual inventory to RFID automation, improving tracking and security. IoT integration enhances shelf organization, and mobile apps engage users with features like reservations. Future advancements may include AIdriven analytics and predictive maintenance for even greater efficiency.

Hardware Requirement

Atmega Microcontroller 328 :

This is the central unit that controls the overall operations. It processes inputs from various components (e.g., RFID reader) and provides output to other modules (e.g., LCD display, voice module).

RFID Reader :

RFID Reader: This module reads RFID tags, typically for identification purposes. It communicates with the Atmega 328 microcontroller to provide data from RFID tags, which the microcontroller can then use to determine actions based on preprogrammed logic.

LCD Display 16x2 :

LCD Display (16×2): A 16×2 LCD screen is used to show information such as user prompts, status updates, or data processed by the microcontroller. This display is directly connected to the Atmega 328.

Power Supply :

This provides power to the entire system, ensuring that each module receives the necessary voltage for operation.

Voice Module This modu le allows for voice-based feedback or instructions. It can be programmed to produce specific voice prompts or alerts based on commands from the microcontroller.

Software Requirement

PYTHON IDLE :

PYTHON IDLE (Integrated Development and Learning Environment) is an ideal tool for the development of the Shelf Smart Library Management System using RFID tags and IoT integration due to its simplicity, versatility, and comprehensive support for various Python libraries and frameworks. Python IDLE provides a user-friendly interface where developers can write, execute, and debug Python code efficiently. Its interactive shell and script editor are well-suited for both rapid prototyping

1. HTML (Hypertext Markup Language) :
HTML is the backbone of any web application, used for structuring content on the web page. It provides the basic layout and elements, such as forms, buttons, and links, that users interact with. In your project:

- Structure: HTML elements will form the structure for your web pages, making them accessible and navigable.
- Forms and Input Fields: Essential for capturing user data, especially in secure login and signup areas.
- Tags and Elements: Define the hierarchy and organization of information in your application.

2. CSS (Cascading Style Sheets) :
CSS is used to style HTML elements, making the web application visually appealing and user-friendly. It will control the layout, color scheme, fonts, and overall aesthetics of your application. Key contributions in your project include:

- Design and Layout: CSS will handle layout adjustments and positioning to make the interface intuitive.
- Responsive Design: Ensures your application is accessible on different devices, including mobile and desktop.

SYSTEM DESIGN

Block Diagram

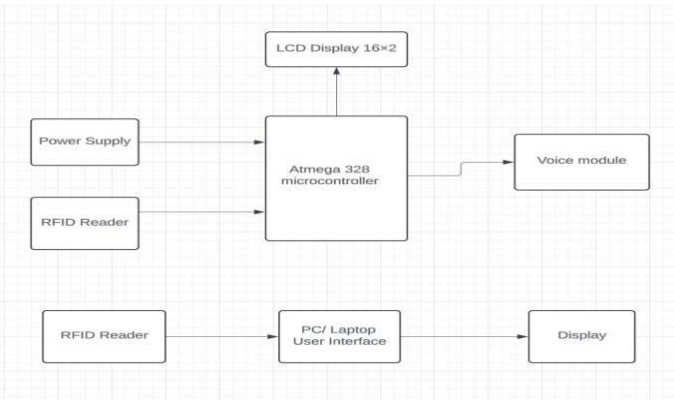

Figure: Block Diagram

SUMMARY

Summary:

The **Shelf Smart Library Management System using RFID tags and IoT integration** marks a bold leap into the future of library management, fusing cutting-edge RFID and IoT technologies to create a seamless, intelligent experience. By automating routine tasks and integrating real-time tracking, smart shelving, and usercentric features, the system transforms traditional libraries into interactive, data-driven hubs of knowledge. Python IDLE facilitates the agile development and testing of this sophisticated system, ensuring reliability and adaptability. This forward-thinking approach not only enhances efficiency and security but also redefines user engagement, laying the foundation for libraries that are smarter, more dynamic, and equipped to meet the evolving needs of modern learners and readers.

REFERENCES

[1] S. Balakrishnan, P. Hemalatha, S. Gridharan, "A Study on Smart Library Management System", AJAST, Vol. 2, Issue 1, January 2018.

[2] H.C. Bansode, S. M. Mhatre, A. N. Patil, G.C. Talkar, "Smart Library using NFC", IJARCET, Vol. 5, Issue 5, May 2016.

[3] Chethan J. Jadhav, Shivani S. Jadhav, "Smart Library Management System using RFID Technology", IRJET, Vol. 4, Issue 5, May 2017.

[4] B. T. Geetha, V. Pereumal, S. Hariharan, T, Jayakumar, "A Smart Book accessing system using RFID and IR in library", IJPAM, Vol. 4, Issue 24, May 2018.

[5] R. Dinesh, S.R. Arun Pravin, M. Aravindhan, D. Rajeshwari, "Library Access system Smartphone Application using Android", IJCSMS, Vol. 4, Issue 1, January 2013.

[6] Santhosh AbajiKharat, B. M. Pange, "QR Codes and academic libraries", IJLIS, Vol. 2, Issue 1, January 2013.

[7] Pooja Shejwal, Amita Wankhade, Vivek Shimpi, Ajit Wale, "A survey on existing barcodes and barcode generation technique", IJCSIT, Vol. 7, Issue 5, July 2016.

[8] Mandeep Kaur, Manjeet Sandhu, Neeraj Mohan, Pavinder S Sandhu, "RFID Technology principles, advantage, limitation, and its application", IJCEE, Vol. 3, Issue 1, February 2011.

[9] Patel R. Integrating AI, GSM, and RFID technologies for intelligent library management systems. Journal of Information Technology Integration. 2023; 8(2): 87-102.

[10] Smith J, Johnson R. Advancements in AI applications for smart libraries.
Journal of Library Automation. 2023; 45(3): 78-84.